

Conditional Probability Calibration for the Five AI-Bubble Stress Cases

Extension of the AI Unwind Stress Framework

August 2026

Abstract

The existing AI unwind framework defines five deterministic stress cases—Mild, Base, Severe, Extreme, and Worst—but does not attach probabilities to them. This note adds a probabilistic layer without attempting to forecast whether or when the AI bubble bursts. The object of interest is therefore the *conditional* distribution of burst severity, $F(D \mid B = 1)$, where $B = 1$ denotes a material bubble crash and D is peak-to-trough drawdown severity. Historical burst severities are taken from the 21 U.S. industry crash episodes reported by Greenwood, Shleifer, and You (2019), whose crash definition is a drawdown of at least 40% within two years after a qualifying price run-up. The five existing model cases are treated as fixed representative points of a continuous severity distribution. Probability mass is assigned to them using Voronoi cells in log-loss space, matching the native additive geometry of the stress framework. Because only 21 burst observations are available, the conditional severity CDF is estimated by a small-sample model ensemble rather than by a single fragile tail distribution. Four plausible bounded/transformed models are fitted by maximum likelihood and combined using AICc weights; uncertainty is assessed with 5,000 episode-level bootstrap replications and cross-checked with a Jeffreys-prior Dirichlet-multinomial estimator. The resulting conditional case probabilities are approximately **Mild 1.3%, Base 46.5%, Severe 37.6%, Extreme 12.5%, Worst 2.1%**. These probabilities sum to 100% *conditional on a burst* and should not be interpreted as estimates of the unconditional probability that the AI bubble will crash.

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1 Executive interpretation

The five stress cases should be interpreted as a discrete approximation to an underlying continuous distribution of bubble-burst severities. A realized crash will almost surely not equal one of the five scenario drawdowns exactly. The correct probability question is therefore not “what is the probability of an exact 53.7% drawdown?” but rather:

$$\mathbb{P}(S = s \mid B = 1)$$

where S is the representative scenario to which the continuous realized severity is assigned, and $B = 1$ means that the bubble has in fact burst.

This conditioning is intentional. The existing stress framework is a severity engine, not a market-timing model. Estimating $\mathbb{P}(B = 1)$ would require a separate hazard model for if and when the current AI bubble breaks. Greenwood, Shleifer, and You (2019) show that run-up characteristics can inform crash likelihood, but this document deliberately does not use that result to time the current market. Instead, it asks a narrower and more defensible question: *given that the bubble has burst, how should probability mass be distributed across the five pre-existing stress cases?*

Conditional scenario-probability calibration: theory pipeline

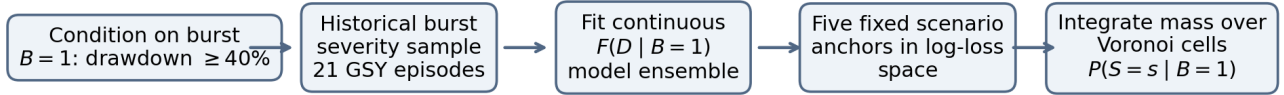


Figure 1: End-to-end logic of the conditional probability extension.

2 Connection to the existing five-case stress framework

2.1 Existing scenario mechanics

The existing model decomposes the AI epicenter shock into a broad-market component and an AI/TMT excess component. In log-return space,

$$S_E^s = S_W^s + \kappa_s S_X^{\text{dotcom}},$$

where:

- S_E^s is the AI epicenter log shock in scenario s ;
- S_W^s is the broad-market log shock;
- S_X^{dotcom} is the full dot-com AI/TMT-excess reference shock;

- κ_s is the severity multiplier attached to scenario s .

The current five-point severity ladder uses approximately $\kappa = 0.335, 0.671, 1.000, 1.500, 2.000$ for Mild through Worst. The Mild case also uses half of the broad-market terminal shock; Base and above use the full broad-market leg.

The resulting representative terminal AI-epicenter drawdowns are:

Scenario	Mild	Base	Severe	Extreme	Worst
κ_s	0.335	0.671	1.000	1.500	2.000
Representative drawdown d_s	31.9%	53.7%	63.9%	75.3%	83.1%
Positive log loss L_s	0.385	0.769	1.019	1.398	1.776

2.2 Why probabilities should be attached to severity regions, not exact points

For a continuous severity random variable D ,

$$\mathbb{P}(D = d_s \mid B = 1) = 0.$$

Therefore the five stress cases are treated as *representative points* rather than literal discrete states of nature. This is the same mathematical idea used in scenario reduction and quantization: a continuous probability distribution is represented by a small finite support, and each support point receives the probability mass of the region it represents. Dupačová, Gröwe-Kuska, and Römisch (2003) develop this idea in stochastic programming using probability metrics; Voronoi-cell scenario construction is a closely related implementation.

3 Conditioning event: what exactly does “the bubble bursts” mean?

Let B denote the burst event. The baseline definition follows Greenwood, Shleifer, and You (2019):

$$B = 1 \iff D \geq 40\%,$$

where D is the maximum peak-to-trough drawdown within 24 months after a qualifying bubble-like price run-up. Their U.S. sample identifies 40 large industry run-ups from 1928–2012, of which 21 subsequently satisfy the 40% crash criterion. The average maximal drawdown among those 21 crashes is approximately 60%.

The conditional severity distribution in this document is thus

$$F_B(d) = \mathbb{P}(D \leq d \mid D \geq 0.40, \text{ qualifying run-up}).$$

The model explicitly *does not* estimate $\mathbb{P}(B = 1)$ for the current AI cycle. If an unconditional scenario probability were ever required, it would need a separate burst-hazard model:

$$\mathbb{P}(S = s) = \mathbb{P}(B = 1) \mathbb{P}(S = s \mid B = 1).$$

The second term is estimated here; the first term is deliberately left outside the scope of the stress framework.

4 Historical conditional severity sample

The primary sample consists of the 21 U.S. industry crash episodes in Table 1, Panel A of Greenwood, Shleifer, and You (2019). The reported maximal 24-month drawdowns range from 44% to 75%, with mean 60.3% and median 60.0%.

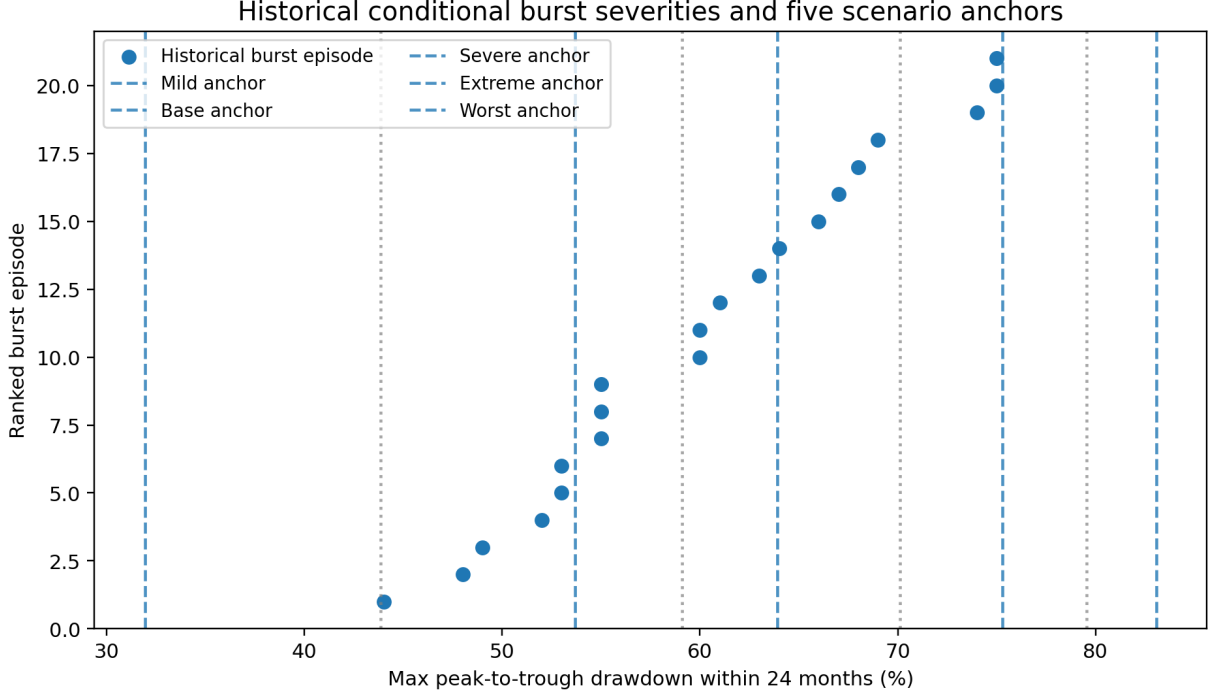


Figure 2: Historical crash severities relative to the existing five AI-unwind scenario anchors. Dotted vertical lines denote Voronoi boundaries.

Two external facts support using these episodes as a severity anchor. First, Greenwood, Shleifer, and You report an average 65% maximal drawdown among their international crash episodes, close to but somewhat more severe than the U.S. 60% average. Second, their dot-com-era U.S. TMT crash observations—Computer Hardware, Computer Software, Electronic Equipment, and Measuring and Control Equipment—average 66.0%. The current AI episode is technologically closer to that subset than to the full all-industry sample, so the all-industry estimate should not be viewed as an aggressively conservative tail calibration.

5 Severity metric and scenario geometry

5.1 Use positive log loss

The stress engine is additive in log-return space, so define the positive log-loss magnitude

$$L = -\log(1 - D).$$

This transformation has three useful properties:

1. L is monotone in simple drawdown D ;
2. it maps $D \in [0, 1)$ to $L \in [0, \infty)$;

3. it is exactly the magnitude corresponding to a negative log-return shock, matching the stress framework.

The 40% burst threshold is therefore

$$u_L = -\log(1 - 0.40) = 0.5108.$$

5.2 Fixed-scenario Voronoi partition

Let $L_1 < \dots < L_5$ be the five fixed representative scenario losses. Assign a realized severity to the closest scenario in squared log-loss distance. The boundary between adjacent scenarios s and $s + 1$ is therefore

$$b_s^L = \frac{L_s + L_{s+1}}{2},$$

and in simple-drawdown terms

$$b_s^D = 1 - e^{-b_s^L}.$$

This produces conditional severity cells:

$$\begin{aligned} C_1 &= [0.40, b_1^D), \\ C_2 &= [b_1^D, b_2^D), \\ C_3 &= [b_2^D, b_3^D), \\ C_4 &= [b_3^D, b_4^D), \\ C_5 &= [b_4^D, 1). \end{aligned}$$

Numerically the boundaries are approximately 43.8%, 59.1%, 70.1%, and 79.5%.

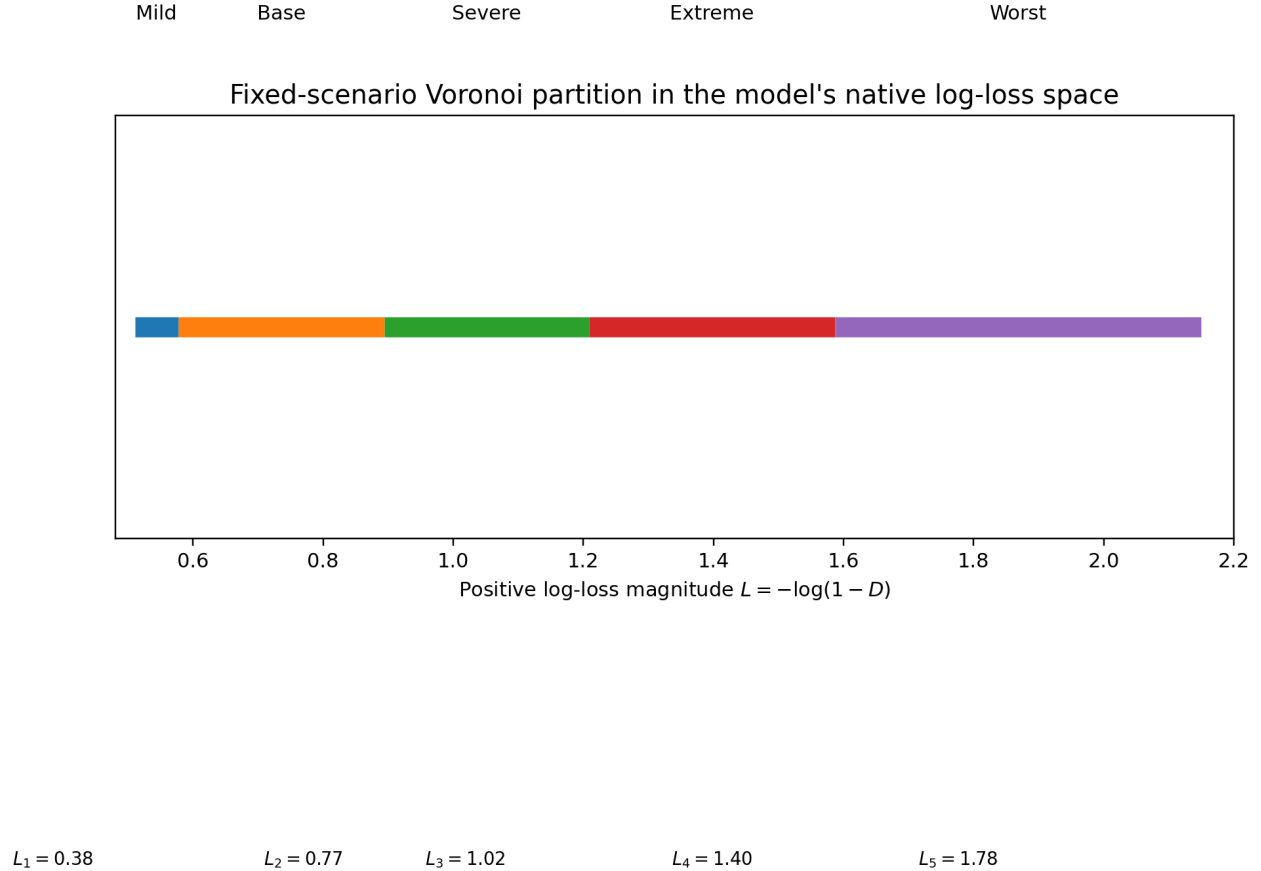


Figure 3: The five stress cases as fixed representative points with Voronoi cells in positive log-loss space.

An important consequence is that the Mild representative anchor itself lies below the 40% conditioning threshold. Conditional on a burst, Mild therefore means “the least severe burst region,” approximately 40–43.8%, rather than a claim that the realized drawdown must equal the original 31.9% Mild anchor. This is mathematically consistent because the scenarios are representatives, not exact states.

6 Why a single extreme-value distribution is not sufficient here

Extreme Value Theory (EVT) provides the natural first theoretical benchmark. Under the Pickands–Balkema–de Haan result, exceedances over a sufficiently high threshold can often be approximated by a Generalized Pareto Distribution (GPD). EVT has a long history in financial tail-risk measurement, including McNeil and Frey (2000), and Bayesian GPD estimation is well developed, for example Castellanos and Cabras (2007).

However, the present application has only 21 event-level crash observations. A free two-parameter GPD fit to excess log losses places its estimated upper endpoint at approximately 75.0% drawdown, essentially at the largest observed historical crash. This is a classic small-sample endpoint instability and would mechanically assign zero probability to the current Worst region above 79.5%. Imposing the natural 100% drawdown endpoint converts the excess log-loss model into a one-parameter exponential tail; a parametric-bootstrap Kolmogorov–Smirnov diagnostic gives $p \approx 0.020$, indicating that this restriction is too strong for the observed conditional sample.

Therefore the final estimator uses EVT as a *theoretical tail motivation and diagnostic*, but deliberately

avoids letting one unstable GPD fit determine the five case probabilities. Instead, it estimates the conditional severity CDF with a small-sample model ensemble.

7 Conditional severity model ensemble

Let

$$Z = \frac{D - u}{1 - u} \in [0, 1], \quad T = L - u_L = \log \left(\frac{1 - u}{1 - D} \right) \in [0, \infty).$$

Four two-parameter candidate models are fitted:

1. $Z \sim \text{Beta}(\alpha, \beta)$;
2. $T \sim \text{Gamma}(k, \theta)$;
3. $T \sim \text{Weibull}(k, \lambda)$;
4. $T \sim \text{Lognormal}(\mu, \sigma)$.

All likelihoods are evaluated in the same D -space, including the appropriate Jacobian for transformed models, so information criteria are comparable.

Because $n = 21$ is small relative to the number of fitted parameters, model selection uses the small-sample corrected Akaike information criterion

$$\text{AICc}_m = -2\ell_m + 2k_m + \frac{2k_m(k_m + 1)}{n - k_m - 1}.$$

The model weight is

$$w_m = \frac{\exp[-\frac{1}{2}(\text{AICc}_m - \text{AICc}_{\min})]}{\sum_j \exp[-\frac{1}{2}(\text{AICc}_j - \text{AICc}_{\min})]}.$$

The model-averaged conditional CDF is

$$F^*(d \mid B = 1) = \sum_m w_m F_m(d \mid B = 1).$$

Finally, the probability attached to scenario s is

$$p_s = \mathbb{P}(S = s \mid B = 1) = F^*(u_s) - F^*(\ell_s),$$

where $[\ell_s, u_s)$ is the Voronoi cell for scenario s .

Model	AICc	Weight	Mild	Base	Severe	Extreme	Worst
Beta on normalized drawdown	-38.38	27.1%	1.1%	47.6%	36.8%	12.3%	2.3%
Gamma on excess log-loss	-38.40	27.5%	1.1%	47.5%	36.9%	12.3%	2.3%
Weibull on excess log-loss	-38.94	35.8%	1.9%	43.3%	40.5%	13.2%	1.1%
Lognormal on excess log-loss	-36.32	9.7%	0.4%	52.0%	31.6%	11.3%	4.7%

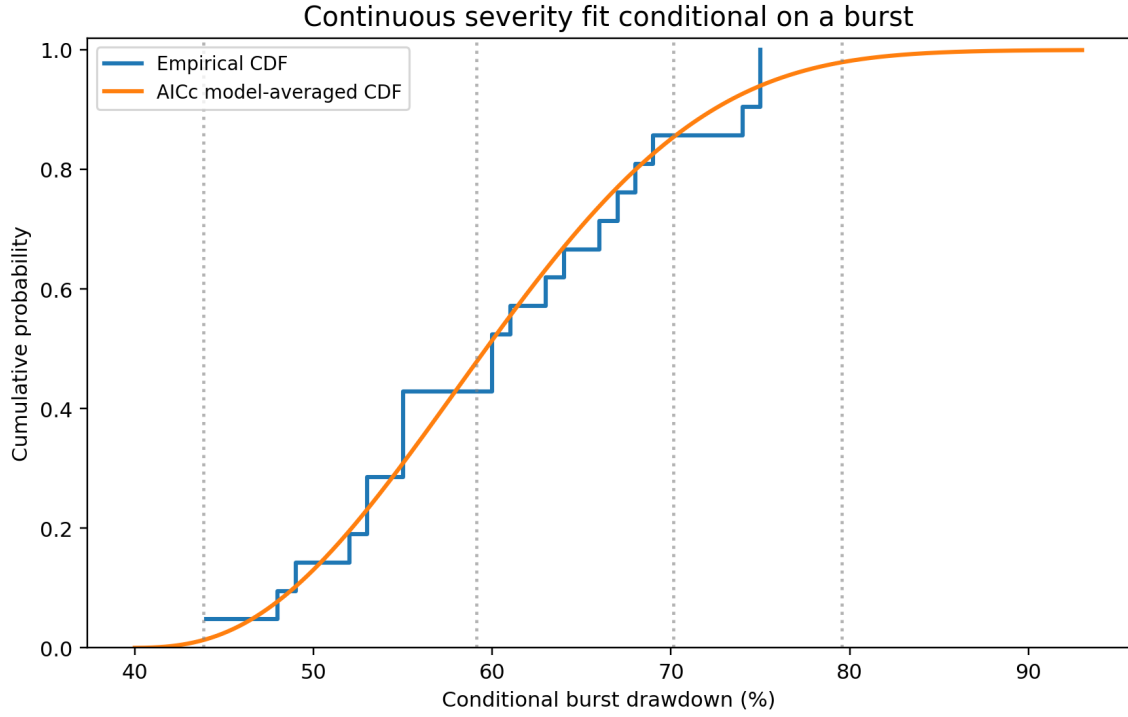


Figure 4: Empirical CDF and model-averaged conditional severity CDF.

8 Sampling uncertainty: episode-level bootstrap

The historical unit of observation is a bubble-burst episode, not a daily return. To preserve that structure, uncertainty is estimated by resampling the 21 burst episodes with replacement. For each of 5,000 bootstrap samples the entire procedure is repeated: each candidate model is re-fitted, AICc weights are recomputed, and the five cell probabilities are re-estimated. This captures both sampling variation and part of the functional-form selection uncertainty.

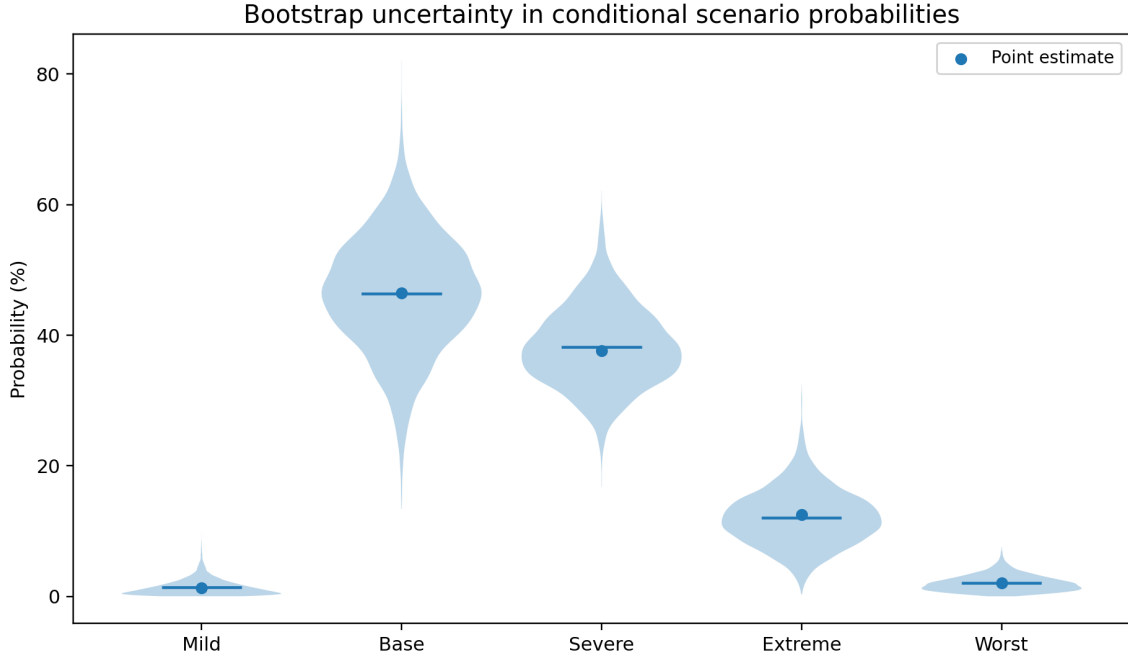


Figure 5: Bootstrap distributions of the five conditional scenario weights.

9 Primary result

The recommended conditional probabilities are:

Scenario	Representative drawdown	Assigned burst region	Probability	90% bootstrap interval	Historical count
Mild	31.9%	40.0–43.8%	1.3%	[0.1%, 3.7%]	0
Base	53.7%	43.8–59.1%	46.5%	[31.8%, 59.9%]	9
Severe	63.9%	59.1–70.1%	37.6%	[28.7%, 48.5%]	9
Extreme	75.3%	70.1–79.5%	12.5%	[5.5%, 19.3%]	3
Worst	83.1%	79.5–100.0%	2.1%	[0.4%, 4.3%]	0

Thus, conditional on a bubble burst, the central estimate is

$$(p_{\text{Mild}}, p_{\text{Base}}, p_{\text{Severe}}, p_{\text{Extreme}}, p_{\text{Worst}}) = (1.3\%, 46.5\%, 37.6\%, 12.5\%, 2.1\%).$$

For management communication, the natural rounded version is approximately

$$(1\%, 47\%, 38\%, 12\%, 2\%).$$

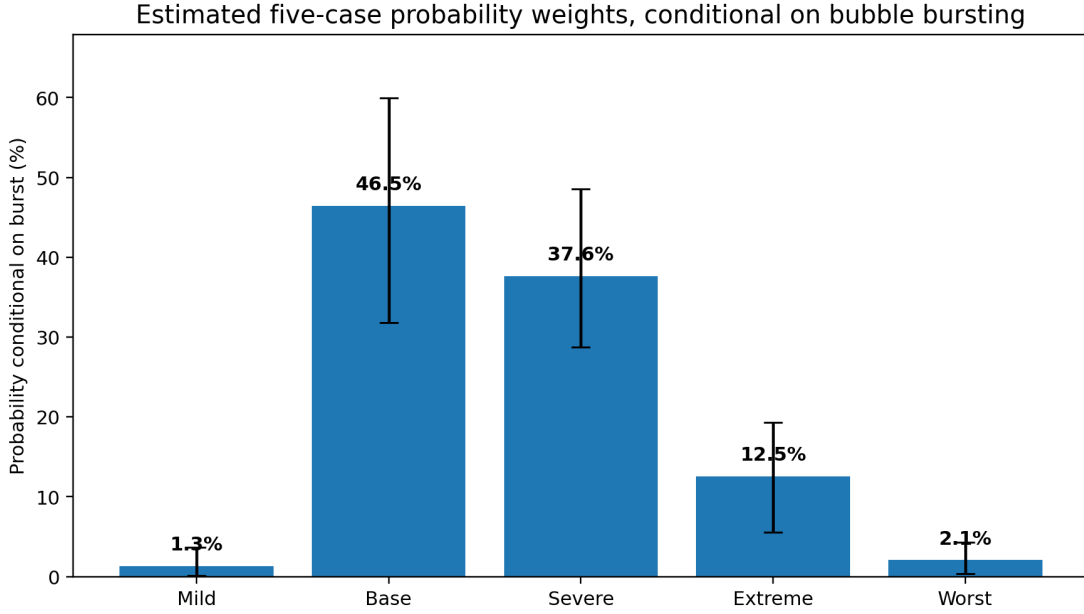


Figure 6: Final five-case probability weights conditional on a burst. Error bars show 5th–95th percentile bootstrap ranges.

9.1 Interpretation

The result puts approximately 84% of conditional probability in the Base and Severe cases. This is intuitive because the historical conditional severity distribution is centered near 60%, between the Base representative drawdown (53.7%) and Severe (63.9%). Extreme receives meaningful but secondary mass, while Mild and Worst are genuine tail states once the conditioning event is already a 40% or larger crash.

The weighted expected drawdown of the five-case approximation is

$$\sum_s p_s d_s = 60.6\%,$$

compared with an observed historical crash-sample mean of 60.3%. In log-loss space the weighted five-case mean is 0.958, compared with the historical mean 0.950. This close match is an important validation: the discrete scenario approximation preserves the central severity of the underlying burst sample rather than merely producing convenient probabilities.

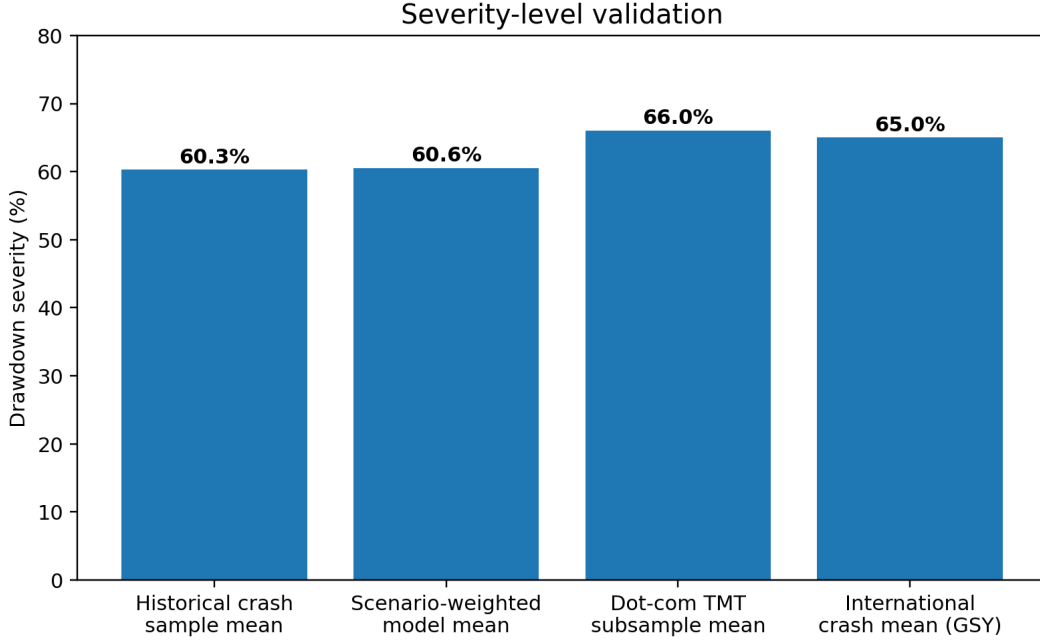


Figure 7: Severity-level validation. The international crash mean is reported by Greenwood, Shleifer, and You (2019); individual international crash episodes are not pooled into the primary likelihood because the published table reports country-level means rather than event-level drawdowns.

10 Non-parametric Bayesian cross-check

To ensure that the result is not an artifact of the four chosen continuous distributions, we also estimate the five case weights directly from historical bin counts. The 21 observed burst episodes fall into the five Voronoi cells with counts

$$(n_1, n_2, n_3, n_4, n_5) = (0, 9, 9, 3, 0).$$

Use the symmetric Jeffreys prior for a multinomial,

$$(p_1, \dots, p_5) \sim \text{Dirichlet} \left(\frac{1}{2}, \dots, \frac{1}{2} \right).$$

The posterior is

$$(p_1, \dots, p_5) \mid \mathcal{D} \sim \text{Dirichlet} \left(n_1 + \frac{1}{2}, \dots, n_5 + \frac{1}{2} \right),$$

with posterior mean

$$\mathbb{E}[p_s \mid \mathcal{D}] = \frac{n_s + 1/2}{n + 5/2}.$$

This yields approximately

$$(2.1\%, 40.4\%, 40.4\%, 14.9\%, 2.1\%),$$

which is close to the continuous model-averaged estimate despite making almost no distributional assumptions.

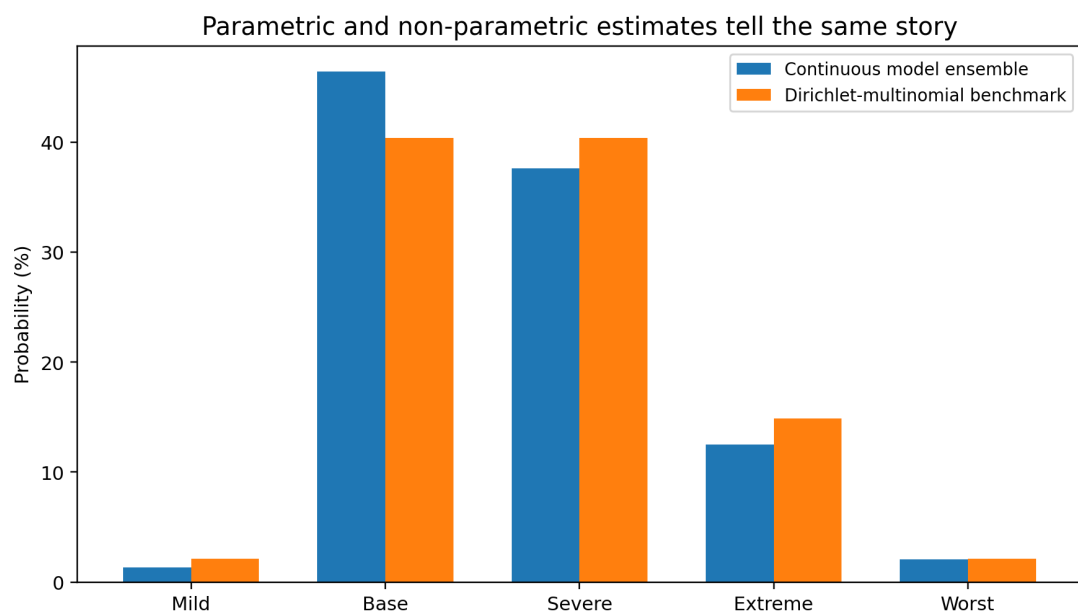


Figure 8: Continuous model ensemble versus a Jeffreys-prior Dirichlet-multinomial benchmark.

11 Functional-form robustness

No single two-parameter family dominates decisively. The Beta, Gamma, and Weibull specifications receive similar AICc support, while the lognormal receives less but non-negligible weight. This is precisely why model averaging is preferable to declaring one tail law to be “the true distribution” with only 21 burst episodes.

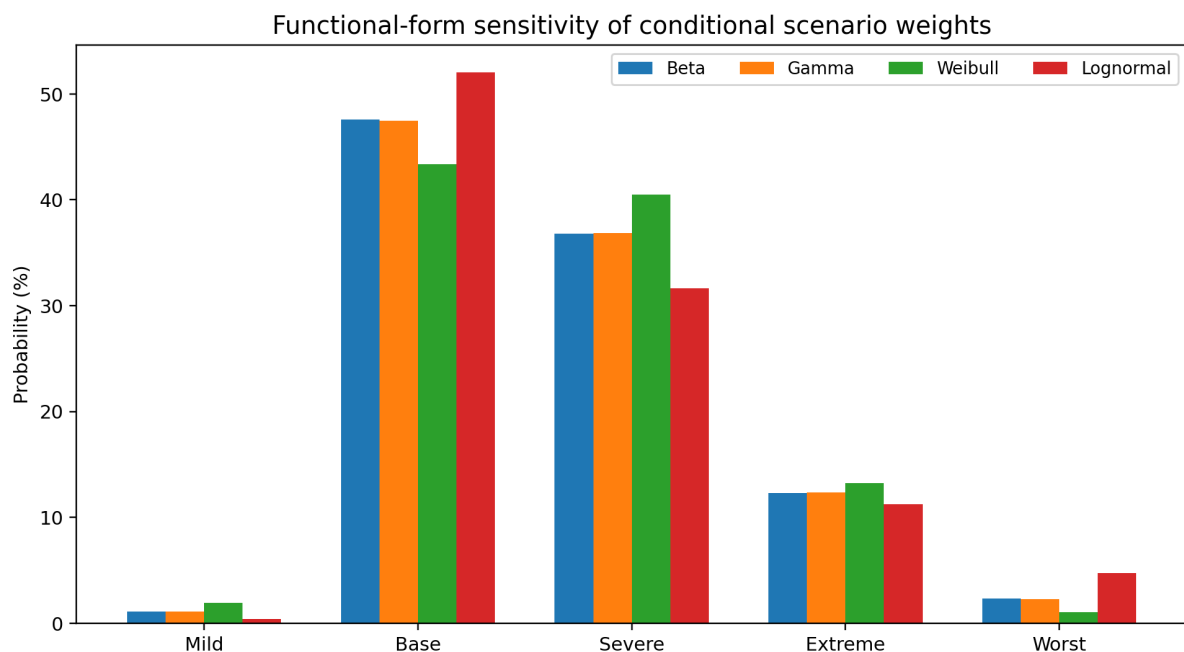


Figure 9: Scenario probabilities under each candidate conditional severity distribution.

The common conclusion across specifications is stable: Base and Severe dominate; Extreme is a meaningful secondary outcome; Mild and Worst are low-probability conditional tails. The exact split between Base and Severe is the main source of model uncertainty.

12 How this probability layer integrates with the existing stress engine

The probability calibration does not alter any existing scenario shock. It adds a weight to each existing case. For any position or portfolio metric Y_s produced by scenario s , the conditional probability-weighted expectation is

$$\mathbb{E}[Y \mid B = 1] = \sum_{s=1}^5 p_s Y_s.$$

For example, if PnL_s is the portfolio stress P&L under scenario s , then

$$\mathbb{E}[\text{PnL} \mid B = 1] = \sum_s p_s \text{PnL}_s.$$

This should be interpreted as a *conditional severity-weighted stress measure*, not as ordinary expected P&L through time. It is meaningful only under the counterfactual conditioning event that the AI bubble has already burst.

Likewise, the five probabilities can be combined with the horizon-specific stress vectors already produced by the model without estimating a different probability distribution at every horizon. The latent random variable is the terminal severity class; the 1w, 2w, 1m, and 2m shocks are the horizon-specific paths associated with that same class. Thus

$$\mathbb{P}(S = s \mid B = 1)$$

is shared across horizons, while the shock magnitude $Y_{s,h}$ changes with horizon h .

13 Why the conditioning approach is preferable to trying to forecast the burst date

There are three conceptual advantages.

1. **Separation of timing and severity.** The probability that a bubble bursts next month is a different problem from the distribution of loss once a burst occurs. Mixing them makes the stress framework dependent on fragile market-timing assumptions.
2. **Consistency with stress testing.** Stress tests are conditional counterfactuals. The user asks “if this adverse state occurs, what happens to the portfolio?” Probability-weighting the severity cases conditional on the state is coherent with that purpose.
3. **Empirical identifiability.** Historical crash severities are directly observable after the fact. A real-time burst hazard is much harder to identify and highly sensitive to how a bubble is defined.

A future extension could separately estimate $\mathbb{P}(B = 1 \mid X)$ using run-up characteristics such as volatility, turnover, issuance, valuation, or price-path acceleration, as suggested by Greenwood, Shleifer, and You. That hazard model should remain a separate module rather than being embedded into the current severity calibration.

14 Limitations and governance

1. **Small event sample.** There are only 21 exact U.S. event-level burst severities in the primary published table. The bootstrap intervals are therefore intentionally wide.
2. **Cross-industry rather than AI-only history.** The sample includes bubbles outside technology. The dot-com TMT subset is more severe on average (66.0%), but is too small to estimate a standalone distribution.
3. **International information is summarized.** The international sample has a reported average crash drawdown of roughly 65%, but the published table gives country-level averages rather than all individual event severities. We therefore use it only as an external check, not as pseudo-observations.
4. **Scenario points are fixed by the existing model.** A fully optimal quantizer would jointly optimize support points and weights. Here the five support points are governance choices already embedded in the stress engine, so only their probabilities are estimated.
5. **Conditional, not unconditional.** The reported numbers do not imply that a 2% Worst probability applies over any calendar horizon. It means that, *if a qualifying burst occurs*, roughly 2% of conditional severity mass maps to the Worst representative region under the baseline estimator.

15 Recommended production numbers

For use in the current five-case AI unwind framework, the recommended baseline conditional weights are:

Mild	1.3%
Base	46.5%
Severe	37.6%
Extreme	12.5%
Worst	2.1%
Total	100.0%

The corresponding rounded management presentation is 1%/47%/38%/12%/2%. Because the Base-versus-Severe split has the largest statistical uncertainty, portfolio decisions should also be checked under the unweighted Severe and Extreme cases rather than relying only on the probability-weighted average.

A Derivation of the Voronoi boundaries

For adjacent scenario points $L_s < L_{s+1}$, the boundary is the severity b satisfying equal squared distance:

$$(b - L_s)^2 = (b - L_{s+1})^2.$$

Expanding and cancelling b^2 gives

$$-2bL_s + L_s^2 = -2bL_{s+1} + L_{s+1}^2,$$

so

$$2b(L_{s+1} - L_s) = (L_{s+1} - L_s)(L_{s+1} + L_s),$$

and therefore

$$b_s^L = \frac{L_s + L_{s+1}}{2}.$$

Mapping back to simple drawdown gives $b_s^D = 1 - e^{-b_s^L}$.

B Historical event table used in the likelihood

Industry	Run-up date	Max drawdown	Assigned case
Automobiles and trucks	Mar 1928	53%	Base
Chemicals	Jan 1929	67%	Severe
Electrical equipment	Jan 1929	55%	Base
Utilities	Jul 1929	53%	Base
Machinery	May 1936	55%	Base
Steel works	Feb 1937	63%	Severe
Tobacco products	Nov 1961	44%	Base
Real estate	May 1968	52%	Base
Personal services	May 1968	69%	Severe
Entertainment	May 1972	60%	Severe
Restaurants and hotels	Jun 1972	55%	Base
Precious metals	Dec 1979	48%	Base
Petroleum and natural gas	Oct 1980	49%	Base
Construction	Oct 1980	64%	Severe
Computer hardware	Mar 1999	68%	Severe
Computer software	Mar 1999	60%	Severe
Electronic equipment	Dec 1999	75%	Extreme
Steel works	Aug 2000	66%	Severe
Measuring and control equipment	Feb 2000	61%	Severe
Steel works	May 2007	75%	Extreme
Coal	Jun 2008	74%	Extreme

C Reproducibility

The accompanying bundle includes:

- the extracted 21-event historical severity CSV;
- the five scenario anchors and probability estimates;
- candidate-model comparison and AICc weights;
- all 5,000 bootstrap probability draws;
- the Python build script used to reproduce the tables, figures, and PDF;
- all figure assets and this LaTeX source.

D References

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